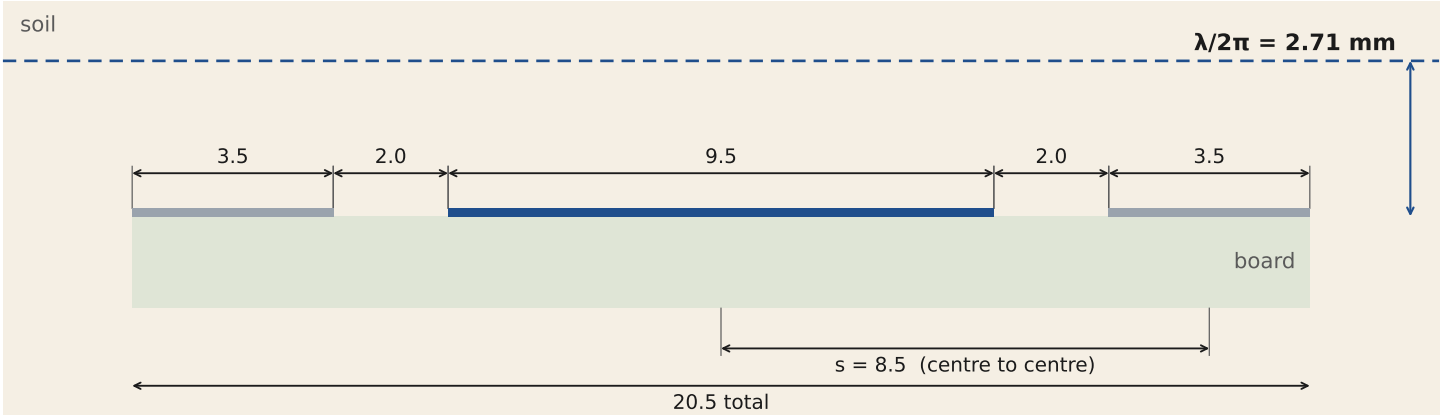
Electrode cross-sections: v1.2, MSV3, MSV4

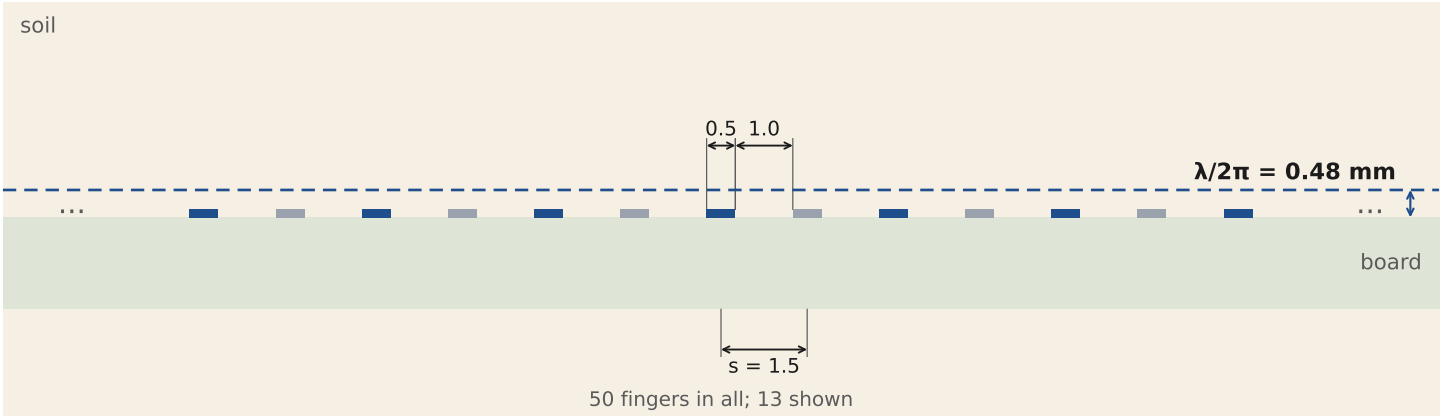
Section across the electrodes, all three at the same scale. Dimensions in mm. Dashed line: 1/e field depth λ/2π.

driven | ground / guard | board | soil

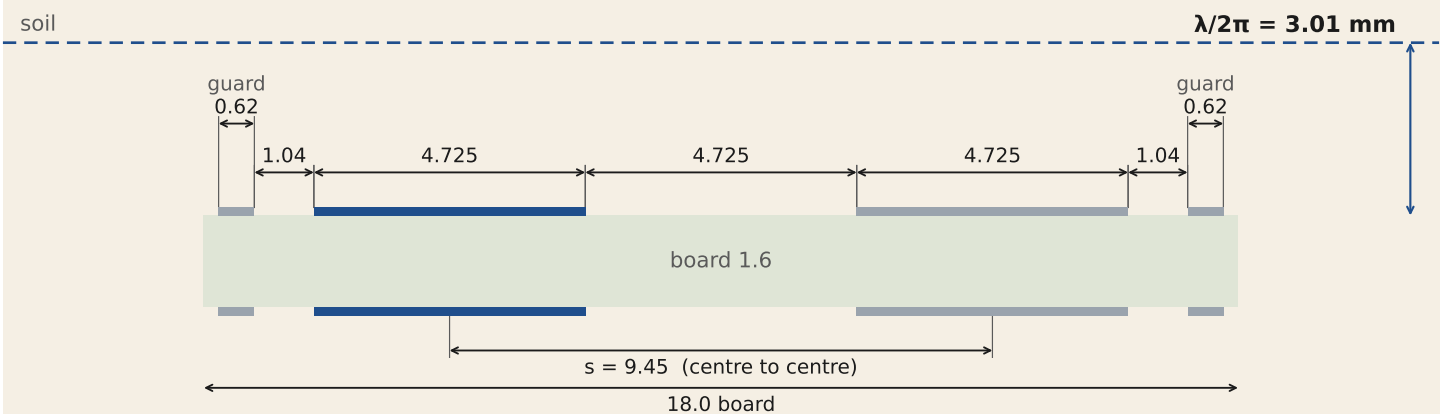
v1.2 — ground | driven | ground, single-sided



MSV3 — interdigitated comb, section across the fingers



MSV4 as drawn — A | B plates inside a guard ring, mirrored front and back



electrode	pattern (mm)	s, centre to centre (mm)	λ = 2s (mm)	λ/2π (mm)
v1.2	G3.5 / 2.0 / H9.5 / 2.0 / G3.5	4.75 + 2.0 + 1.75 = 8.5	17.0	**2.71**
MSV3	0.5 finger / 1.0 gap, 50 fingers	0.25 + 1.0 + 0.25 = 1.5	3.0	**0.48**
MSV4 as drawn	A4.725 / 4.725 / B4.725	2.3625 + 4.725 + 2.3625 = 9.45	18.9	**3.01**

λ is the spatial period of the pattern: twice the driven-to-ground centre spacing s. λ/2π is the 1/e depth of the field for an ideal periodic array. The 2D field solve puts the median signal depth at 1.39 mm (v1.2), 0.23 mm (MSV3) and 1.01 mm (MSV4); MSV4's guard ring makes it shallower than λ/2π suggests. Board thickness is drawn at 1.6 mm for all three (only MSV4's is recorded); copper thickness is exaggerated.